

Samsung bada coming in 2010

Samsung Electronics has unveiled its open mobile platform, Samsung bada, which will enable developers to create applications for new Samsung mobile phones.

'bada,' which means ocean in Korean, was chosen to convey the limitless variety of potential applications which can be created using the new platform. bada OS, according to Samsung, also represents the fresh challenges and opportunities available to developers, as well as the entertainment which consumers will enjoy once the new platform is open.

The first bada-powered mobile phone is expected to be introduced in the first half of 2010. Multiple models of bada phones will be launched in the second half of next year. Sale of bada applications will start at the Samsung Application Store in the same period.

32GB NAND flash chip

Samsung has reduced the size of its flash memory by packing 32GB into a chip only 0.05cm thick. The new 'ultra-thinning' technology allows a significant increase in storage for smartphones and other mobile devices.

It overcomes the conventional technology that limits a chip's resistance to external pressure when under 30 micrometres (μm) in height. As a result, now 32GB NAND flash chips can be created, each measuring only 15 μm .

The new technology will deliver memory that is 40 per cent thinner and



Samsung 32GB NAND flash chips

lighter and provide handset designers with much greater freedom to create attractive designs.

Intel, NEC push boundaries of supercomputing

Intel and NEC Corp. plan to jointly develop high-performance computing (HPC) system technologies to push the boundaries of supercomputing.

NEC will bring these technologies to market in future supercomputers based on the Intel Xeon processor. NEC's expertise in this field, coupled with the Intel Xeon processor's performance and its accelerating vector capabilities such as Advanced Vector Extensions (AVX), will allow for higher performance supercomputers.

Carbon nanotube fibres

Researchers at Rice University have made long fibres by processing carbon nanotubes in a solution called super-acid. These long fibres can be used as lightweight, efficient wires for the electrical grid, or as the basis of structural materials and conductive textiles.

Individual carbon nanotubes are strong, lightweight and electrically conductive, and could be valuable as, amongst other things, electrical transmission wires. But aligning masses of the nanotubes into well-ordered materials such as fibres has been a challenge for manufacturers. The researchers used acid processing methods to assemble carbon nanotubes into fibres 50 micrometres thick and hundreds of metres long.

Durable silicon-air battery

Scientists at Technion-Israel Institute of Technology have developed a new silicon-air battery that is capable of working non-stop thousands of hours.

Silicon is a more common material. It is more stable, light-weight and has high energy capacity (four electrons are transferred during oxidation of a single silicon atom).

Scientists actually turned it into sand (silicon dioxide) during battery usage. This will be a non-rechargeable battery like similar batteries around today.

Scientist converts iPhone into chemical sniffer

A NASA scientist has developed a postage-stamp-sized chemical sensor that allows iPhones to sniff out low



The chemical sensing prototype plugged into an iPhone 30-pin dock connector with the display-side up. (Photo credit: Dominic Hart/NASA)

airborne concentrations of chemicals such as ammonia, chlorine gas and methane.

The gadget add-on was developed by Jing Li, a physical scientist at NASA's Ames Research Centre in California. She developed the proof of concept with other researchers as part of Homeland Security's 'Cell-All' programme, which aims to put more mobile sensors in the hands of cellphone users.

High-performance 3-axis MEMS gyroscope

STMicroelectronics has introduced a high-performance single-package